

# Auction-Dynamics MBO on Surface Meshes

The threshold-dynamics replacement for Phase 1: provenance, derivation, what transfers from the literature, and what does not

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## Abstract

Approach B (`src/partition/mbo.auction.py`) replaces the Phase 1  $\Gamma$ -convergence relaxation by volume-constrained multiphase threshold dynamics: the state is a hard labelling of the mesh vertices, each step diffuses every cell's indicator by one backward-Euler step of the surface heat equation and reassigns every vertex by a balanced thresholding  $\arg \max_k [y_{ik} + \psi_k]$  whose offsets  $\psi$  enforce equal discrete area. None of the ingredients is new: the scheme is Merriman–Bence–Osher's, its variational interpretation is Esedoğlu–Otto's, the balanced thresholding is Jacobs–Merkurjev–Esedoğlu's auction dynamics, and the implicit-Euler diffusion is the one the graph-MBO literature uses. This document first records, from the papers themselves, what each of those works did, at what cell counts, on what domains — because an earlier description of this approach overstated its novelty on three of its four clauses. It then derives what is specific to this implementation: that the backward-Euler step is an *exponential mixture* of heat kernels, from which its interface constant ( $\frac{1}{2}$ , against  $1/\sqrt{\pi}$  for the Gaussian) and its per-step displacement ( $\tau\kappa/2$ , against  $\tau\kappa$ ) follow in three lines each and are confirmed on the production mesh; that the balanced threshold is the dual of a transportation problem whose two-phase case is the Ruuth–Wetton shifted level; that the Esedoğlu–Otto dissipation theorem does *not* apply to the implemented pairing of a consistent diffusion with a lumped assignment and an inexact dual, and exactly which inequality survives; that a lumped–lumped pairing would restore the theorem (not implemented); and the two-sided time-step window, with the over-merge side quantified by the diffused indicator's value at a cell's centre,  $1 - xK_1(x)$  at  $x = R_{\text{cell}}/\sqrt{\tau}$ . Measurements are cited from report 08; nothing here is a new measurement except the one-step displacement check.

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# 1 Overview

## Provenance of every computed number in this document

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status              | derivation, complete; twice adversarially reviewed (2026-09-17), all findings applied                                                                                                                                                                                                                                                                                                                                                                        |
| Script              | <code>docs/math/10-mbo-auction-dynamics/check_numerics.py</code> → committed <code>numerics.yaml</code> beside this file. <b>Every number below that was produced by running code is quoted from that file</b> , with its key named in the text; nothing is quoted from an interactive session or from a reviewer’s scratch script. Run from the repo root ( $\approx 5$ min; <code>--quick</code> skips the two $100 \times 96$ dense eigendecompositions). |
| Determinism         | jittered disc centres (§5.6): <code>numpy.random.default_rng(0)</code> , 24 samples per row, one shared generator consumed row by row in the table’s order, offsets $\theta \sim U(0, 2\pi/348)$ , $\varphi \sim U(0, 2\pi/328)$ . Labelling pair (§7.2): the geodesic balanced init at the project seed 84172851, then one full-budget balanced step.                                                                                                       |
| Meshes              | torus $R = 1.0$ , $r = 0.6$ throughout ( <code>TorusMeshProvider</code> ); $40 \times 36$ , $60 \times 52$ , $100 \times 96$ (the production base), and the $348 \times 328$ deliverable mesh ( $V = 114,144$ ); $\tau$ from <code>tau_diagnostics</code> with <code>MBOConfig()</code> defaults.                                                                                                                                                            |
| Versions            | Python 3.12.13, numpy 2.4.4, scipy 1.17.1; macOS 15.3.1 arm64 (Mac mini). Eigenvalues may differ in the last printed digit across BLAS builds; nothing quoted depends on that digit.                                                                                                                                                                                                                                                                         |
| Cited, not computed | the run statistics of §7.2 and §7.3 (4 of 96 energy increases; the slack range) are from <code>docs/experiments/08-mbo-auction-dynamics/data.yaml</code> ; the 68.16%/1.97% of §9 from <code>docs/plans/PHASE1_BC.REPLACEMENT.PLAN.md</code> ; the freeze constants’ $c = 2/c = 4$ pair from <code>testing/test_mbo_auction.py</code> ; every statement about a paper from the copy indexed in <code>docs/papers/README.md</code> .                          |

Phase 1 of this project, as specified by Bogosel and Oudet [5] and derived in document 06 (`docs/math/06-phase1-energy-discretization/`), minimises a Modica–Mortola functional over  $N$  continuous densities under an equal-mass constraint, and reads the partition off as the winner-take-all territory of those densities. Document 09 (`docs/math/09-balanced-readout/`) derives why that readout can diverge from what was constrained — at high  $N$  it manufactures cells whose territory is far from the equal-area target, and cells whose territory is disconnected — and the extraction-time correction that repairs it.

Approach B removes the category of problem rather than repairing it. Its state is a labelling  $\omega : \{1, \dots, V\} \rightarrow \{1, \dots, N\}$  of the mesh vertices, never a continuous field, so there is no readout to diverge. Each step at a fixed mesh level is

1. **diffuse**: for every cell  $k$ , solve  $(M + \tau K) y_k = M \chi_k$ , where  $\chi_k$  is the indicator of cell  $k$ ’s territory;
2. **balanced threshold**:  $\omega'(i) = \arg \max_k [y_{ik} + \psi_k]$ , with the  $N$  offsets  $\psi$  chosen so that every cell’s discrete area equals the target  $\bar{A}$ .

This is repeated until the labels stop changing, the labels are carried to the next finer mesh by nearest-vertex interpolation, and the process repeats. Phase 2 then refines the resulting contour exactly as it would a Phase 1 output (`src/optimization/perimeter_optimizer.py`); nothing downstream changed.

The measurements — perimeter against the validated PGD deliverables, the three validity gates, attribution to the dynamics rather than the initialisation, seed spread, and the  $N = 400$ – $1000$  runs — are in `docs/experiments/08-mbo-auction-dynamics/` and are not repeated here. This document is the derivation and the provenance. Following the scope policy of `docs/math/AUTHORING_GUIDE.md`, it derives only what `src/partition/mbo_auction.py` computes; where a cleaner variant exists on paper (§7.4) it is stated and marked as not implemented.

## 2 Provenance: what was done before, and what is claimed here

Every reference in this section was read from the copy held in docs/papers/ (title page, abstract, and the sections cited), not from a secondary description.

### 2.1 The lineage

**Threshold dynamics.** Merriman, Bence and Osher introduced the scheme in a 1992 UCLA report [22]: diffuse the indicator of a set for a short time, threshold at  $\frac{1}{2}$ , repeat; the boundary moves by mean curvature. Their 1994 paper [23] extends it to several phases by diffusing every indicator and assigning each point to the largest — the  $\arg \max$  step above. Convergence of the two-phase scheme to mean-curvature flow was proved by Evans [15]; the multiphase case, in the sense of a BV weak formulation and conditionally on energy convergence, by Laux and Otto [19]. Ruuth [26] extended the multiphase scheme to prescribed junction angles.

**The variational interpretation.** Esedoğlu and Otto [14] showed that one MBO step is the minimiser of the *linearisation* of a “heat content” energy  $E_\tau$  at the current configuration, that this energy is concave on the convex hull of partitions, so the step never increases it (their Lemma 5.2, for any kernel  $G \geq 0$  with  $\hat{G} \geq 0$ , §5.2), and that  $E_\tau$   $\Gamma$ -converges to the total interfacial energy as  $\tau \rightarrow 0$  (their Appendix A, for smooth radial kernels). Esedoğlu and Jacobs [13] widen the  $\Gamma$ -convergence to kernels that are merely nonnegative, symmetric, integrable with a finite first moment, and positive near the origin (their Theorem 5.1), which is the class the present kernel belongs to (§5.4). The underlying nonlocal-to-local  $\Gamma$ -limit theory — the convergence of a nonlocal perimeter approximation to the surface-tension model, which [14, ref. 1] invokes for “heat content tends to perimeter” — is Alberti and Bellettini [2].

**Volume constraints.** Ruuth and Wetton [27] enforce a prescribed volume in the two-phase scheme by thresholding the diffused indicator not at  $\frac{1}{2}$  but at the level  $l$  that preserves the phase volume (their Algorithm, step 2(b)); [20] proves the scheme’s convergence. Jacobs, Merkurjev and Esedoğlu [18] generalise this to  $N$  phases: minimising the linearisation of  $E_\tau$  under volume constraints is a linear assignment problem, its Lagrange multipliers shift the thresholds (their eq. (11), which is exactly the  $\arg \max_k [y_{ik} + \psi_k]$  of this implementation), and they solve it with Bertsekas’ auction algorithm [4]; hence “auction dynamics”. Laux and Swartz [20] prove convergence of the volume-preserving thresholding scheme.

**Graphs, and the implicit-Euler diffusion.** Garcia-Cardona et al. [16] run multiclass MBO on a weighted graph and replace the Gaussian convolution by an *implicit-Euler* step of the graph heat equation,  $U^{n+1/2} = B^{-1}[U^n - \delta t \mu(U^n - \hat{U})]$  with  $B = I + \delta t L_s$  (their eqs. (27)–(28); eq. (29) is  $B$  in a truncated eigenbasis). Jacobs et al. [18, §5] run *volume-constrained* auction dynamics on an unstructured  $k$ -nearest-neighbour graph with unit-volume nodes — but with the symmetrised similarity matrix  $W^\top W$  used *directly* as the kernel, no Laplacian and no implicit step. Van Gennip, Guillen, Osting and Bertozzi [28] prove the graph scheme is *pinned* for  $\tau$  below a spectral bound and *trivial* above another (their Theorems 4.2–4.4) — the rigorous form of the two-sided window of §8.

**Surfaces.** Merriman and Ruuth [24] move curves with junctions on a curved surface by extending each indicator off the surface with the closest-point map, diffusing in  $\mathbb{R}^3$  on a Cartesian grid, and thresholding on the surface; their §8.3 evolves five regions on an embedded torus and, in their Fig. 11, the 174 countries plus ocean of a world map on the sphere — many-region multiphase threshold dynamics on a curved closed surface, unconstrained. Wang and Osting [30] compute *Dirichlet* partitions (minimal sum of first Laplace–Beltrami eigenvalues, not perimeter) of flat tori and of the sphere by a diffusion-generated method, via FFT and the spherical harmonic

transform, for  $k \leq 20$ . The surface finite-element setting used here — P1 mass and stiffness matrices on a triangulation of an embedded surface — is Dziuk’s [9, 10].

**Partitions with many cells.** Elsey, Esedoğlu and Smereka simulate grain growth by diffusion-generated motion with 166,927 grains in two dimensions [11] and 133,110 in three [12], on uniform grids, without volume constraints. Wang et al. [29] apply iterative convolution–thresholding to piecewise-constant Mumford–Shah segmentation with a handful of phases; Hu, Liu and Wang [17] use auction dynamics as the inner solver of the *outer* (Pólya) problem on  $256^2$  and  $128^3$  grids with  $n \leq 9$  cells. Bogosel and Oudet [6] initialise the same outer problem from capacity-constrained Voronoi diagrams with more than 100 equal-area cells in the plane.

| Work      | Domain, discretisation                         | Cells                             | Equal-area                   | Relation to this implementation              |
|-----------|------------------------------------------------|-----------------------------------|------------------------------|----------------------------------------------|
| [18] §4.2 | flat 2-torus $[0, 1]^2$ , grid, FFT            | 17, <b>64</b> ; 160               | <b>yes</b> (160: pre-served) | same step, same problem class, exact auction |
| [18] §4.2 | flat 3-torus                                   | 8; 32                             | yes (32: pre-served)         | Weaire–Phelan found with temperature         |
| [24] §8.3 | embedded torus and sphere, closest point, grid | 5 (torus); 175 (sphere)           | no                           | multiphase MBO on curved closed surfaces     |
| [30]      | flat tori and $S^2$ , FFT / SHT                | $\leq 20$                         | no                           | other energy; curved closed surface          |
| [17]      | box, $256^2$ / $128^3$ grid                    | $\leq 9$                          | yes                          | auction dynamics as inner solver             |
| [16]      | weighted graph, eigenbasis                     | $\sim 10$ classes                 | no                           | implicit-Euler diffusion step                |
| [18] §5   | unstructured $k$ NN graph, kernel $W^\top W$   | $\sim 10$ classes                 | bounds                       | constrained + unstructured, no surface       |
| [11, 12]  | grids                                          | $10^5$                            | no                           | scale, unconstrained                         |
| [6]       | plane                                          | $> 100$ (init only)               | yes                          | capacity-constrained Voronoi as init         |
| [5]       | meshed surfaces, $\Gamma$ -convergence         | $\leq 11$ torus, $\leq 32$ sphere | yes                          | the Phase 1 incumbent; no thresholding       |

Table 1: Cell counts and settings, read from the cited sections. “Preserved” means each Voronoi cell keeps its *initial* area or volume (160 cells in 2D, 32 grains in 3D), not that they are equal; “init only” means the count appears in an initialisation figure, not a minimisation run.

## 2.2 The corrected statement of what is new

An earlier description of approach B (the module docstring, and §2 of report 08 as first written) said that every published implementation of this machinery runs “on a uniform Cartesian grid via FFT, on a flat domain, at small  $N$ , for Pólya’s outer problem”. Table 1 falsifies three of those four clauses: [18] §4.2 solves the *inner* problem at  $N = 64$  and 160 (two clauses), and [24] runs the multiphase scheme on a curved torus and on the sphere (the “flat” clause; JME’s torus is flat). Only “uniform grid via FFT” survives, and only for the Euclidean implementations; the graph line never used an FFT.

What the held literature does *not* contain, stated as “we are not aware of” and not as “first”:

1. the scheme on a **curved, embedded surface discretised by an unstructured triangle mesh**, with the diffusion done by a surface-FEM backward-Euler step — the graph line has the operator form, the closest-point line has the surface, neither has both, and neither works with lumped and consistent mass on a triangulation;
2. **equal-area constraints at  $N = 400$ – $1000$** : in the held papers, equal-area minimisation reaches 64 cells [18, §4.2]; the 160 and 32 there are area-preserving, and the “more than 100” of [6] is an initialisation figure;

3. the assignment solved by an **inexact** transportation dual (§6) rather than an exact auction, together with the inequality that survives inexactness (§7.3);
4. the **multilevel mesh ladder** with label interpolation and a per-level, mesh-derived time step (§10, §8);
5. coupling to an exact contour refinement (Phase 2) and the gate instruments of report 07.

The defensible framing is therefore *combination, surface, and scale*, not *method*. The method is [18]’s; the diffusion operator is the graph-MBO one transplanted to surface finite elements; “large  $N$ ” by itself is not new in three separate lines of work. And the held papers come closer than a one-line summary suggests: [18, §5] already has the volume constraint on an *unstructured* domain (a similarity graph), [24, Fig. 11] has 175 regions on a curved closed surface, and [30] has a curved closed surface with exact spectral diffusion at 20 cells. What none of them has is all of it at once on a triangulated surface with equal-area constraints in the hundreds.

### 3 Notation and Setup

The surface  $S$  is closed and triangulated with vertices  $\mathbf{V} = \{x_1, \dots, x_V\}$  and faces  $\mathbf{F}$ .  $M$  and  $K$  are the P1 mass and stiffness matrices of docs/math/06-phase1-energy-discretization/eq. (MK-def), assembled by `TriMesh.compute_matrices`;  $K$  is the cotangent Laplacian and  $K\mathbf{1} = 0$ . The lumped mass is  $v_i = \sum_j M_{ij} > 0$ ,  $D = \text{diag}(v)$ ,  $|S| = \sum_i v_i$ , and the equal-area target is  $\bar{A} = |S|/N$ . The mesh scales are the mean and maximum edge length  $h_{\text{mean}}$ ,  $h_{\text{max}}$  over all triangle sides, and the cell scale is the radius of the equal-area disc,

$$R_{\text{cell}} = \sqrt{\bar{A}/\pi}. \quad (1)$$

A labelling  $\omega$  has indicator columns  $\chi_k \in \{0, 1\}^V$ ,  $\chi_{ik} = [\omega(i) = k]$ , so  $\sum_k \chi_k = \mathbf{1}$ , and territories  $\Omega_k = \{i : \omega(i) = k\}$  with discrete areas  $T_k = \sum_{i \in \Omega_k} v_i = v^\top \chi_k$ . The diffused indicators are the columns  $y_k$  of  $y \in \mathbb{R}^{V \times N}$ . In the continuum,  $G_t$  is the heat kernel of  $S$  at time  $t$  (on  $\mathbb{R}^2$ ,  $G_t(x) = (4\pi t)^{-1} e^{-|x|^2/4t}$ ), and  $K_\tau^{\text{res}}$  the kernel of  $(I - \tau \Delta)^{-1}$ , defined in §5.4.

Code: the objects

```
src/partition/mbo-auction.py:      sparse_one_hot(labels, n.cells)      builds       $\chi$ ;
edge_lengths(mesh) gives  $h_{\text{mean}}$ ,  $h_{\text{max}}$ ; cell_radius(total_area, n.cells) is (1); mesh.v is  $v$ ,
mesh.M, mesh.K are  $M$ ,  $K$ .
```

## 4 Threshold dynamics in the continuum

### 4.1 The MBO scheme

For a set  $\Sigma \subset S$  and a time step  $\tau$ , one step of [22] is

$$\Sigma' = \{x : (G_\tau * \chi_\Sigma)(x) \geq \tfrac{1}{2}\}, \quad (2)$$

and for a partition  $(\Omega_1, \dots, \Omega_N)$  one step of [23] is

$$\omega'(x) = \arg \max_k (G_\tau * \chi_{\Omega_k})(x). \quad (3)$$

Since  $\sum_k G_\tau * \chi_{\Omega_k} = 1$ , (3) reduces to (2) when  $N = 2$ . Away from junctions, each interface moves by mean curvature over time  $\tau$  per step, to leading order in  $\tau$  [15, 14]; §5.5 derives the constant for the kernel actually used here.

## 4.2 The heat-content energy and the linearisation principle

Following [14, §4], define for a partition

$$E_\tau(\omega) = \frac{1}{\sqrt{\tau}} \sum_{k=1}^N \int_S \chi_{\Omega_k} (1 - G_\tau * \chi_{\Omega_k}) dA = \frac{1}{\sqrt{\tau}} \sum_{k \neq j} \int_S \chi_{\Omega_k} G_\tau * \chi_{\Omega_j} dA, \quad (4)$$

the second form by  $\sum_j G_\tau * \chi_{\Omega_j} = 1$ . Each term measures the heat that escapes from cell  $k$  into the others in time  $\tau$  — “heat content”. Note the sum is over *ordered* pairs, so every interface is counted from both sides; this is eq. (4.2) of [14] with all surface tensions equal to one and the same  $\tau^{-1/2}$  normalisation.

Relaxed to  $u : S \rightarrow \Delta_N$  (the probability simplex), the energy is

$$E_\tau(u) = \frac{1}{\sqrt{\tau}} \left[ |S| - \sum_k \int_S u_k G_\tau * u_k dA \right], \quad (5)$$

which is **concave**:  $u_k \mapsto \int u_k G_\tau * u_k$  is a quadratic form with a positive semi-definite kernel ( $\widehat{G_\tau} = e^{-\tau|\xi|^2} \geq 0$ ), so its negative is concave. Its linearisation at  $u$  is, up to a constant,

$$L_u(u') = -\frac{2}{\sqrt{\tau}} \sum_k \int_S u'_k G_\tau * u_k dA, \quad (6)$$

and minimising  $L_u$  over simplex-valued  $u'$  is a pointwise problem whose solution puts all of  $u'(x)$  on the largest  $(G_\tau * u_k)(x)$  — i.e. exactly the step (3). This is the observation of [14] that makes threshold dynamics a minimising-movements scheme, and it carries the dissipation property:

**Proposition 4.1** (Dissipation; [14, Lemma 5.2]). *Let  $F$  be concave and differentiable on a closed convex set  $\mathcal{K}$ , and let  $u' \in \arg \min_{\mathcal{K}} L_u$  where  $L_u$  is the linearisation of  $F$  at  $u \in \mathcal{K}$ . Then  $F(u') \leq F(u)$ .*

*Proof.* Concavity gives  $F(u') \leq L_u(u')$ ; optimality of  $u'$  and feasibility of  $u$  give  $L_u(u') \leq L_u(u) = F(u)$ .  $\square$

The proof uses only that  $u$  itself lies in  $\mathcal{K}$ . So the conclusion survives *any* further convex constraint on  $u'$  that  $u$  already satisfies — in particular the equal-area constraint of §6, provided the constrained linearisation is minimised *exactly*. That is the route by which [18] obtain dissipation for auction dynamics, and §7 examines which of its hypotheses the implementation violates.

## 4.3 $\Gamma$ -convergence and the interface constant

For a kernel  $K_\tau(x) = \tau^{-d/2} K(x/\sqrt{\tau})$  with  $K \geq 0$ , radial,  $\int K = 1$  and a finite first moment, the heat content of a smooth set satisfies, per unit interface length and to leading order in  $\tau$ ,

$$\frac{1}{\sqrt{\tau}} \int_{\Sigma^c} K_\tau * \chi_\Sigma dA \longrightarrow c_K \text{Per}(\Sigma), \quad c_K = \frac{1}{2} \int_{\mathbb{R}^d} |z_1| K(z) dz = \int_0^\infty r k(r) dr, \quad (7)$$

where  $k$  is the one-dimensional marginal of  $K$ . The second form is the flat calculation: for an interface  $\{x_1 = 0\}$ , the pairs  $(s, t)$  with  $s < 0 < t$  and  $t - s = r$  have measure  $r$ , so the escaped heat per unit length is  $\int_0^\infty r k_\tau(r) dr = \sqrt{\tau} \int_0^\infty r k(r) dr$ . Summed over cells in (4), every interface is counted from both sides, so

$$E_\tau(\omega) \longrightarrow c_K \sum_k \text{Per}(\Omega_k). \quad (8)$$

For the Gaussian,  $k$  is  $N(0, 2)$  and  $c_G = \frac{1}{2}\sqrt{4/\pi} = 1/\sqrt{\pi}$ . This is the constant  $c_0 = (|B^{d-1}|/|S^{d-1}|) \int |z| G(z) dz$  of [14, eq. (A.2)]: in  $d = 2$  it is  $\pi^{-1}\sqrt{\pi} = 1/\sqrt{\pi}$ , and the same value follows from (7) because  $\int |z_1| K = (|B^{d-1}|/|S^{d-1}|) \cdot 2 \int |z| K$  for a radial kernel. The rigorous statement —  $\Gamma$ -convergence of (5) in  $L^1$  to  $c_K \sum_k \text{Per}(\Omega_k)$ , with compactness — is [14, Proposition A.1] for kernels that are smooth, nonnegative, radial, radially non-increasing, of unit mass with a finite first moment, and satisfy  $|\nabla G(z)| \lesssim G(z/2)$ ; and [13, Theorem 5.1] for kernels that are merely nonnegative, symmetric, in  $L^1$  with  $|z|K \in L^1$ , and positive in a neighbourhood of the origin. The kernel used here fails the first set (it is singular at the origin, §5.4) and satisfies the second. Both theorems are proved on a periodic box in  $\mathbb{R}^d$ ; neither covers a curved surface or a finite-element discretisation. That transfer is assumed here, not proved.

## 5 The surface finite-element diffusion step

### 5.1 One backward-Euler step

The semi-discrete P1 heat equation on the surface is  $M\dot{y} + Ky = 0$  [10]. One backward-Euler step of length  $\tau$  from  $y(0) = \chi_k$  is

$$(M + \tau K) y_k = M \chi_k, \quad \text{i.e.} \quad y_k = A_\tau \chi_k, \quad A_\tau = (M + \tau K)^{-1} M. \quad (9)$$

This is the only linear solve in the method.  $M + \tau K$  is symmetric positive definite, is factorised once per (level,  $\tau$ ) pass — in the step loop, again for the probe baseline and each probe factor, and for the final full-budget assignment — and the  $N$  right-hand sides are back-substituted.

Code: the diffusion step

```
src/partition/mbo-auction.py: make_diffusion_solver(mesh, tau) returns factorized((M
+ tau*K).tocsc()) (SuperLU); diffuse_indicators(solve, mesh, labels, n_cells) forms
Mχ sparsely and solves column by column, accumulating  $\sum_k (M\chi_k)^\top y_k$  for §11.
```

### 5.2 The step is an exponential mixture of heat semigroups

Everything that is specific to this kernel follows from one identity. Let  $L = M^{-1}K$ ; it is diagonalisable with real eigenvalues  $0 = \lambda_1 \leq \lambda_2 \leq \dots$  by the generalised eigenproblem  $K\phi = \lambda M\phi$ , and  $e^{-tL}$  is the semi-discrete heat semigroup.

**Proposition 5.1** (Mixture representation).

$$A_\tau = (I + \tau L)^{-1} = \int_0^\infty \frac{1}{\tau} e^{-t/\tau} e^{-tL} dt, \quad (10)$$

*i.e. one backward-Euler step is the heat semigroup averaged over an exponentially distributed time  $t \sim \text{Exp}(\text{mean } \tau)$ . The same identity holds in the continuum,  $(I - \tau \Delta)^{-1} = \int_0^\infty \tau^{-1} e^{-t/\tau} e^{t\Delta} dt$ .*

*Proof.* On each eigenvector,  $\int_0^\infty \tau^{-1} e^{-t/\tau} e^{-t\lambda} dt = \tau^{-1}(\tau^{-1} + \lambda)^{-1} = (1 + \tau\lambda)^{-1}$ ; the eigenvectors span  $\mathbb{R}^V$ .  $\square$

### 5.3 Structural properties

**Proposition 5.2.** (i) *Partition of unity.*  $\sum_k y_k = \mathbf{1}$  exactly, for every labelling.

(ii) *Symmetry.*  $MA_\tau = M(M + \tau K)^{-1}M$  is symmetric positive definite, so  $\chi \mapsto -\chi^\top MA_\tau \chi$  is concave.

(iii) **Positivity is not guaranteed on the mesh.** In the continuum  $A_\tau$  has a positive kernel (§5.4). On a triangulation,  $K$  has positive off-diagonal entries across obtuse angles and  $M$  is not diagonal, so  $e^{-tL}$  need not be entrywise nonnegative (on a  $40 \times 36$  torus mesh its minimum entry is  $-1.4 \times 10^{-2}$  at  $t = 0.1h^2$ ), and no argument gives nonnegativity of the mixture  $A_\tau$  either. In practice it holds: on every torus mesh tested at the production  $\tau$  the minimum entry of  $A_\tau$  is positive ( $+2.7 \times 10^{-5}$  at  $40 \times 36$ ,  $N = 10$ ;  $+8.7 \times 10^{-9}$  at  $60 \times 52$ ,  $N = 100$ ;  $+3 \times 10^{-10}$  and 0 to twelve digits at  $100 \times 96$ ,  $N = 100$  and 400), so the caveat is inactive on the ladder. (numerics.yaml: positivity\_and\_pairing.exp\_m\_check, pairing\_cases[\*].min\_entry\_A\_tau.)

*Proof.* (i)  $K\mathbf{1} = 0$  gives  $(M + \tau K)\mathbf{1} = M\mathbf{1}$ , so  $A_\tau\mathbf{1} = \mathbf{1}$ , and  $\sum_k \chi_k = \mathbf{1}$ . (ii)  $(M + \tau K)^{-1}$  is SPD and  $M$  is symmetric, so  $M(M + \tau K)^{-1}M$  is SPD. (iii) is the standard failure of the discrete maximum principle for the cotangent Laplacian on non-Delaunay meshes, and the numbers are direct computation. Nothing in the method relies on positivity: the assignment is invariant under adding a common constant to a vertex's scores, and the balanced threshold does not compare against  $\frac{1}{2}$ .  $\square$

## 5.4 The continuum kernel and its interface constant

By (10) the kernel of  $(I - \tau\Delta)^{-1}$  on  $\mathbb{R}^2$  is

$$K_\tau^{\text{res}}(r) = \int_0^\infty \frac{1}{\tau} e^{-t/\tau} \frac{e^{-r^2/4t}}{4\pi t} dt = \frac{1}{2\pi\tau} K_0\left(\frac{r}{\sqrt{\tau}}\right), \quad (11)$$

with  $K_0$  the modified Bessel function. From the mixture, without evaluating (11):  $K_\tau^{\text{res}}$  is nonnegative, radial, of unit mass, has all moments  $\int |z|^p K_\tau^{\text{res}} = \mathbb{E}_t[\int |z|^p G_t] \propto \mathbb{E}[t^{p/2}] < \infty$ , and is positive (indeed logarithmically singular) at the origin. It therefore lies in the class of [13, Theorem 5.1]; its Fourier symbol  $(1 + \tau|\xi|^2)^{-1}$  is positive, so it also satisfies the hypothesis of Proposition 4.1's continuum version [14, §5.2]. Its tails are exponential rather than Gaussian: the one-dimensional marginal is

$$k_\tau(s) = \frac{1}{2\sqrt{\tau}} e^{-|s|/\sqrt{\tau}}, \quad (12)$$

because restricting the symbol to the  $\xi_1$ -axis gives  $(1 + \tau\xi_1^2)^{-1}$ , the transform of (12).

**Proposition 5.3** (Interface constant of the backward-Euler kernel).  $c_{K_\tau^{\text{res}}} = \frac{1}{2}$ . Hence, for the kernel actually used,

$$E_\tau(\omega) \longrightarrow \frac{1}{2} \sum_k \text{Per}(\Omega_k) = \text{total interface length, each interface counted once.} \quad (13)$$

*Proof.* Either from (12):  $\int_0^\infty r k_1(r) dr = \frac{1}{2} \int_0^\infty r e^{-r} dr = \frac{1}{2}$ . Or from the mixture: the Gaussian at time  $t$  escapes  $\sqrt{t/\pi}$  per unit length, and  $\mathbb{E}[\sqrt{t}] = \sqrt{\tau} \Gamma(\frac{3}{2}) = \frac{1}{2} \sqrt{\pi\tau}$ , so  $\tau^{-1/2} \mathbb{E}[\sqrt{t/\pi}] = \frac{1}{2}$ .  $\square$

**Remark 5.1** (The factor of two in the perimeter convention). Every perimeter this project reports — `final_perimeter`, `arm_report.yaml`, the deliverable tables — is  $\sum_k \text{Per}(\Omega_k)$ , which counts each interface twice. By (13),  $E_\tau$  approximates *half* of that quantity, i.e. the same thing `label_boundary_length` measures (§11). With the Gaussian it would approximate  $(2/\sqrt{\pi}) \approx 1.128$  times the interface length instead.

## 5.5 The per-step displacement

**Proposition 5.4** (Displacement per step). *Let  $\Sigma$  have a smooth boundary with (geodesic) curvature  $\kappa$  at a point, positive when  $\Sigma$  is locally convex. One step (2) with the Gaussian  $G_\tau$  moves the boundary inward by  $\tau\kappa$ ; with the backward-Euler kernel  $K_\tau^{\text{res}}$  it moves it by*

$$\delta = \frac{\tau\kappa}{2}, \quad (14)$$

*both to leading order in  $\tau$ . One backward-Euler MBO step is therefore mean-curvature flow over time  $\tau/2$ , not  $\tau$ .*

*Proof.* At signed distance  $d$  outside  $\Sigma$ , the Taylor expansion of the diffused indicator is [14, §5]

$$(G_t * \chi_\Sigma)(d) = \frac{1}{2} - \frac{d + t\kappa}{\sqrt{4\pi t}} + O(d^2, t), \quad (15)$$

so the Gaussian threshold set is  $d = -t\kappa$ . Averaging (15) over  $t \sim \text{Exp}(\tau)$  and setting it to  $\frac{1}{2}$ ,

$$d \mathbb{E}[t^{-1/2}] + \kappa \mathbb{E}[t^{1/2}] = 0 \implies d = -\kappa \frac{\mathbb{E}[t^{1/2}]}{\mathbb{E}[t^{-1/2}]} = -\kappa \frac{\sqrt{\tau} \Gamma(3/2)}{\tau^{-1/2} \Gamma(1/2)} = -\frac{\kappa\tau}{2}. \quad \square$$

**Remark 5.2** (Second moments do not determine the speed). Both kernels have second moment  $2\tau$  per coordinate (expand  $(1 + \tau|\xi|^2)^{-1}$  and  $e^{-\tau|\xi|^2}$  to first order), so one might expect the same speed. The displacement is instead  $\delta = \frac{\kappa}{2} \rho_K$  with  $\rho_K = \int y_2^2 K(0, y_2) dy_2 / \int K(0, y_2) dy_2$  the ratio of the transverse second moment to the marginal density *on the interface*;  $\rho_K$  is  $2\tau$  for the Gaussian and  $\tau$  for  $K_\tau^{\text{res}}$  (a direct computation with (11), using  $\int_0^\infty s^2 K_0(s) ds = \int_0^\infty K_0(s) ds = \pi/2$ ). The mixture proof avoids the Bessel integrals. Every identity in §5.2–§5.5 — the mixture against the closed-form kernel, both interface constants, the displacement by both routes, the Bessel integrals, the second moments, and the centre values of §8.3 — is evaluated numerically in `numerics.yaml: constants`.

## 5.6 Numerical check on the production mesh

Proposition 5.4 is for the continuum kernel and to leading order in  $\tau$ ; the implementation uses (9) on a triangulation. For a disc the continuum problem is exactly solvable, which gives a reference that is not leading-order. With  $x = R/\sqrt{\tau}$  and  $u = r/\sqrt{\tau}$ , the radial solution of  $(1 - \tau\Delta)y = \chi_{B_R}$  is

$$y(u) = \begin{cases} 1 - xK_1(x)I_0(u), & u \leq x, \\ xI_1(x)K_0(u), & u \geq x, \end{cases} \quad (16)$$

(continuous with continuous derivative at  $u = x$  by the Wronskian  $I_0K_1 + I_1K_0 = 1/x$ ; the centre value is (24)), so the  $\frac{1}{2}$ -threshold radius  $u^*$  and the exact area loss  $\pi\tau(x^2 - u^{*2})$  follow by a one-dimensional root find. At the  $x$  of the table below it is  $1.09\text{--}1.16 \times \pi\tau$ : the leading-order  $\pi\tau$  is low by 9–16% at these  $x$ , because the resolvent’s exponential tails make its  $O(1/x)$  correction about four times the Gaussian’s (whose exact loss is  $1.02\text{--}1.04 \times 2\pi\tau$  at the same  $x$ ).

On the  $348 \times 328$  torus mesh of the deliverables ( $V = 114,144$ ,  $h_{\text{mean}} = 0.0171$ ,  $h_{\text{max}} = 0.0311$ ), take the set of vertices within *ambient* distance  $R_d$  of a point on the outer equator — not a geodesic disc; its lumped area differs from  $\pi R_d^2$  by  $-1.4$  to  $+0.8\%$  — apply one step of (9) and a  $\frac{1}{2}$ -threshold, and measure the lumped area change. The result depends on where the centre sits relative to the mesh: the boundary ring of vertices flips in groups, so a single placement carries granularity noise. The table gives the vertex-centred placement  $(1.6, 0, 0)$  and the mean over 24 centres drawn uniformly within one grid cell around it, as ratios to the exact continuum loss (16) (all values: `numerics.yaml: disc_check`; seed and draw order in the provenance block):

| $R_d$ | $\tau$ | $\sqrt{\tau}/h_{\text{mean}}$ | $x$  | exact loss | vertex-centred | jittered mean $\pm$ std | vs. Gaussian exact |
|-------|--------|-------------------------------|------|------------|----------------|-------------------------|--------------------|
| 0.20  | 0.004  | 3.69                          | 3.16 | −0.01454   | 0.90           | $0.97 \pm 0.06$         | 0.54               |
| 0.25  | 0.004  | 3.69                          | 3.95 | −0.01392   | 0.89           | $0.96 \pm 0.08$         | 0.52               |
| 0.30  | 0.006  | 4.52                          | 3.87 | −0.02095   | 0.84           | $0.93 \pm 0.04$         | 0.50               |
| 0.20  | 0.002  | 2.61                          | 4.47 | −0.00684   | 0.77           | $0.91 \pm 0.09$         | 0.48               |

Averaged over placement the mesh reaches 0.91–0.97 of the exact continuum loss and 0.48–0.54 of the exact Gaussian value; the vertex-centred placement is the low tail of that distribution (an earlier version of this section reported it alone and read a 10–23% deficit into it). Of the remaining 3–9% shortfall, surface curvature accounts for 1.4–3.1% (the Gaussian curvature at the outer equator is  $1/(r(R+r)) = 1.042$ , which lowers the boundary’s geodesic curvature below  $1/R_d$  by  $KR_d^2/3$ , i.e. 3.1% at  $R_d = 0.3$ ); the rest is finite-element and threshold-granularity error at a per-step displacement of only 0.25–0.63  $h_{\text{mean}}$  (implied by the measured vertex-centred loss, `displacement_over_h_mean`). What the check establishes is that the FEM step moves an interface at *half* the Gaussian rate to within the granularity of the test, as Proposition 5.4 says. This is a one-off check of the derivation, not a measured study.

## 6 The balanced threshold

### 6.1 The constrained linearisation is a transportation problem

Minimising the linearisation (6) under the equal-area constraint, discretised with a weight matrix  $W$  for the inner product, is

$$\max_{\chi'} \sum_k \langle \chi'_k, W y_k \rangle \quad \text{s.t.} \quad \sum_k \chi'_{ik} = 1 \quad \forall i, \quad \sum_i v_i \chi'_{ik} = \bar{A} \quad \forall k, \quad \chi'_{ik} \in \{0, 1\}. \quad (17)$$

The implementation takes  $W = D$ , so the objective is  $\sum_i v_i y_{i, \omega'(i)}$  and (17) is the transportation problem (LP) of docs/math/09-balanced-readout/ §6.2 with score matrix  $s_{ik} = y_{ik}$ : supplies  $v_i$  at the vertices, demands  $\bar{A}$  at the cells. With unit vertex masses it would be the assignment problem of [18]; with the real lumped masses it is a transportation problem [1], which is why exact balance is generically unattainable (doc 09, §6.5) and why a granularity bar exists at all.

### 6.2 Dual, shifted argmax, and the two-phase antecedent

Doc 09 derives the dual of the LP relaxation and shows (its Proposition 6.1) that complementary slackness forces the assignment onto

$$\omega'(i) = \arg \max_k (y_{ik} + \psi_k), \quad (18)$$

in the code’s sign convention, where raising  $\psi_k$  grows cell  $k$ , and that the dual subgradient is  $T_k - \bar{A}$ : grow the cells that are too small. The  $\psi_k$  are the Lagrange multipliers  $\lambda^*$  of [18, eq. (11)] and the *prices* of the auction [4]. For  $N = 2$ , using  $y_{i1} + y_{i2} = 1$ ,

$$y_{i1} + \psi_1 \geq y_{i2} + \psi_2 \iff y_{i1} \geq \frac{1}{2} + \frac{\psi_2 - \psi_1}{2}, \quad (19)$$

which is the threshold level shifted away from  $\frac{1}{2}$  until the volume is met — the scheme of [27]. (18) is its  $N$ -phase form.

### 6.3 The inexact solver as implemented

`solve_dual_offsets` is a subgradient method on the piecewise-linear dual with a decaying step — descent on  $D(\psi)$  in doc 09’s convention, ascent on the concave form in the code’s — returning the *best* iterate visited (doc 09, §6.2–6.3). Three things are specific to its use here:

1. **Scale.** The step schedule was calibrated on log-density scores. Diffused indicators live on  $[0, 1]$ , so with `normalize_scores=True` the scores are divided by the median gap between each vertex's top two scores (the margin an offset must cover to flip a vertex), the schedule `normalized_eta0 = 2.0`, `dual_decay = 0.02`, `normalized_dual_iters = 2000` is run on the normalised scores, and  $\psi$  is rescaled to the caller's units. Correctness of (18) is scale-free; only convergence is not.
2. **Quality bar.** A solve is accepted at  $\max_k |T_k - \bar{A}|/\bar{A} \leq \text{bar}$  with  $\text{bar} = \max(2\gamma, \text{A-parity})$ , where  $\gamma = \max_i v_i/\bar{A}$  is the one-vertex granularity and A-parity is approach A's own stall level at equal budget (report 07). A-parity is a two-entry lookup, for  $(V, N) = (47, 488, 300)$  and  $(114, 144, 300)$  only; for every other configuration — all  $N = 100, 400\text{--}1000$  runs and every non-torus surface — the bar is exactly  $2\gamma$ . It is a solver-failure detector, not a validity test: equal area is what the step optimises, so hitting it says the dual converged.
3. **Early stopping is production-only.** In-loop solves stop at the bar; every level ends with a full-budget re-assignment, and in-loop area figures are tagged `early_stop_capped` so they are never quoted as assignment quality (report 07, artefact 5).

Code: the balanced threshold

```
src/partition/mbo-auction.py:          balanced_assign(scores, lumped_mass, target,
bar, early_stop) → src/partition/balanced-readout.py: solve_dual_offsets with
BalancedReadoutConfig(normalize_scores=True, dual_early_stop_rel=bar or None);
assignment_margin_scale is the median top-two gap; src/partition/arm_harness.py:
assignment_quality_bar, vertex_granularity.
```

**Remark 6.1** (Why an inexact standard solver, and what may be claimed for it). Nothing in §6 is a new method for the transportation dual. Subgradient ascent on that dual — grow the cells that are too small — is the standard solver for this class of problem [3, 21, 25], and doc 09 (its Proposition 6.1) proves that the implemented update *is* that subgradient. What is specific to this work is the *decision* to use it inside the threshold step in place of the exact auction of [18], and the reasons are these, each on record:

1. the adaptation was forced by the mesh regardless of solver — the auction of [18] is stated for unit-volume points, and with real lumped masses the problem is a transportation problem whose optimum is generically fractional (§6, doc 09 §6.5), so a granularity bar is the correct stopping rule for *any* solver;
2. a validated primitive already existed: the routine was built for the balanced readout and qualified through the Phase 0 harness (`docs/experiments/07-phase0-shared-harness/`), which reproduces the reference Phase 2 perimeter bitwise;
3. inexactness is safe *because the scheme is iterated*: report 07 records that the solver looked inadequate only on a cold-start fixture that no iterated method faces, and that from the first step every assignment starts from an already-balanced partition and converges under the bar — a conditional judgment, checked rather than assumed;
4. the cost is bounded, not waived: the dissipation theorem is lost (§7.2), the inequality that replaces it holds for any priced labelling (Proposition 7.1) and its slack is measured on every run (§7.3).

The limit of the claim: the choice *sufficed*, and its price is known. It is **not** established that it is better, faster or more accurate than the auction, because no auction was implemented or measured on this problem; the one alternative that was measured, entropic (Sinkhorn) regularisation [8], lost to the incumbent on five of six score matrices (report 07). This remark exists so that the mathematical record carries the reasoning behind the choice and the disclaimer with it, not only

the consequences.

## 7 Energy dissipation: what transfers and what does not

### 7.1 The theorem, for the consistent pairing with an exact assignment

Discretise (5) with the consistent mass matrix:

$$E_\tau^M(\chi) = \frac{1}{\sqrt{\tau}} \left[ |S| - \sum_k \chi_k^\top M A_\tau \chi_k \right] = \frac{1}{\sqrt{\tau}} \left[ |S| - \sum_k \langle \chi_k, M y_k \rangle \right]. \quad (20)$$

By Proposition 5.2(ii) this is concave on the convex hull of labellings, and its linearisation at  $\chi$  is  $-\frac{2}{\sqrt{\tau}} \sum_k \langle \chi'_k, M y_k \rangle$ . So if the assignment step *exactly* maximised  $\sum_k \langle \chi'_k, M y_k \rangle$  over labellings with equal discrete area, Proposition 4.1 would give  $E_\tau^M(\chi') \leq E_\tau^M(\chi)$  at every step, constraint included — the discrete form of the auction-dynamics theorem of [18].

### 7.2 Why the implementation is outside it

Two hypotheses fail, independently.

1. **The pairing.** The assignment maximises  $\sum_k \langle \chi'_k, D y_k \rangle = \sum_k \chi_k'^\top D A_\tau \chi_k$  with the *lumped*  $D$ , while  $y_k$  was produced with the *consistent*  $M$ , and  $D A_\tau = D(M + \tau K)^{-1} M$  is not symmetric. Write  $D A_\tau = S + Z$  with  $S$  symmetric and  $Z$  skew;  $\|Z\|_F / \|D A_\tau\|_F$  is  $2 \times 10^{-3}$  on the  $40 \times 36$  and  $60 \times 52$  torus meshes and  $0.9\text{--}1.2 \times 10^{-3}$  on the  $100 \times 96$  production base. The lumped energy  $E_\tau^D = \tau^{-1/2} [ |S| - \sum_k \chi_k^\top S \chi_k ]$  (the skew part drops out of a quadratic form) is concave iff  $S \succeq 0$ , which is not guaranteed but was true on every mesh tested at the production  $\tau$ : smallest eigenvalue  $+4 \times 10^{-6}$  ( $40 \times 36$ ,  $N = 10$ ;  $60 \times 52$ ,  $N = 100$ ) and  $+6 \times 10^{-7}$ ,  $+2 \times 10^{-6}$  ( $100 \times 96$  at  $N = 100, 400$ ), against largest eigenvalues of  $3 \times 10^{-3}$  to  $2 \times 10^{-2}$ . The operative failure is elsewhere: the linearisation of  $-\chi^\top S \chi$  at  $\chi$  is  $-2S\chi$ , whereas the assignment maximises  $\langle \chi', (S + Z)\chi \rangle$ . The step is therefore the exact minimise-the-linearisation step of Proposition 4.1 for  $E_\tau^D$  *perturbed by the skew term*  $\langle \chi', Z\chi \rangle$ . The matrix-norm figure above is an upper bound on that perturbation, not its size: at an actual pair of labellings (the geodesic balanced init at the project seed, then one full-budget balanced step)  $\langle \chi', Z\chi \rangle / \langle \chi', S\chi \rangle$  is  $1.1 \times 10^{-5}$  on  $40 \times 36$  and  $2.0 \times 10^{-5}$  on  $60 \times 52$ , and the skew term changes the unconstrained argmax at one vertex of 1,440 and one of 3,120 respectively (`numerics.yaml: pairing_cases[*].labelling_pair`). All eigenvalue and asymmetry figures in this item: `pairing_cases[*]`. Monotonicity of  $E_\tau^D$  can fail only through that term and through inexactness (next item); tracking  $E_\tau^M$  instead is simply the wrong pairing, since its gradient  $-2M A_\tau \chi$  is not what the assignment sees at all.
2. **Exactness.** [18] obtain a theorem because the auction solves the assignment exactly (to  $\varepsilon$ -complementary slackness). `solve_dual_offsets` returns the best iterate of a subgradient method with a residual area error, and the early-stopped in-loop solves are further from optimal by design.

Report 08 records the consequence:  $E_\tau^D$  *increases* on 4 of 96 active steps at  $N = 300$ , consistent with two small perturbations of an otherwise monotone scheme. Neither energy is therefore a gate. The discrete forms of the dissipation statement exist in the literature and are the right reference here: [13, §4.4] state that two-phase graph MBO with a symmetric weight matrix  $W$  decreases its energy at every step when  $W$  is positive semi-definite, and exhibit a  $W = A + \lambda I$  ( $\lambda < 2$ ) for which it cycles between two configurations (their Example 3, Fig. 3); their numbered Propositions 4.7–4.9 concern modified algorithms with split weights and are not the statement used here. For the constrained multiclass scheme [18, §5.2] state the same condition (“as long as

$W$  is positive semi-definite, the graph heat content is concave”), which is Proposition 4.1 applied to a graph energy.

### 7.3 The inequality that survives

**Proposition 7.1** (Descent inequality). *Let  $\omega'$  be any labelling of the form (18) for some  $\psi \in \mathbb{R}^N$  — exact or not — and  $\omega$  the previous labelling. Then*

$$\sum_k \langle \chi'_k, Dy_k \rangle - \sum_k \langle \chi_k, Dy_k \rangle \geq \sum_k \psi_k (T_k - T'_k), \quad (21)$$

where  $T_k, T'_k$  are the old and new discrete areas.

*Proof.* For every vertex  $i$ , the argmax gives  $y_{i\omega'(i)} + \psi_{\omega'(i)} \geq y_{i\omega(i)} + \psi_{\omega(i)}$ . Multiply by  $v_i > 0$  and sum: the  $y$  terms give the left side of (21), and  $\sum_i v_i (\psi_{\omega(i)} - \psi_{\omega'(i)}) = \sum_k \psi_k (T_k - T'_k)$ .  $\square$

The right-hand side is exactly the price of inexactness: at perfect balance  $T_k = T'_k = \bar{A}$  it vanishes and (21) becomes the textbook monotone step for the lumped linearisation; otherwise it scales like  $\|\psi\| \times (\text{area error})$ : the per-level maxima recorded in report 08’s `data.yaml` run from  $1.3 \times 10^{-4}$  ( $N = 100$ ) to  $2.8 \times 10^{-2}$  ( $N = 1000$  on  $V = 114,144$ , level 1), relative to  $|v^\top y_\omega|$ . Because (21) needs neither concavity nor exactness, a violation beyond floating point can only mean the returned labels are not the argmax at the returned  $\psi$ , or a unit mismatch between them; that makes it worth gating on, and the measured violation is  $\leq 6.1 \times 10^{-8}$ .

Code: energies and the gate

```
src/partition/mbo_auction.py:      lyapunov_energy(y, labels, v, total_area, tau,
inner_consistent) returns E_lumped =  $E_\tau^D$  and E_consistent =  $E_\tau^M$ ; descent_slack(y, v,
psi, labels_old, labels_new, n_cells) returns lhs, rhs of (21) and violation_rel =
(rhs-lhs)/scale. testing/test_mbo_auction.py gate G4 asserts (21).
```

### 7.4 A pairing that would restore the theorem (not implemented)

**Remark 7.1** (Lumped–lumped pairing). Replace (9) by the mass-lumped step  $(D + \tau K) y_k = D\chi_k$ , i.e.  $A_\tau^D = (D + \tau K)^{-1}D$ , and keep the assignment as it is. Then  $DA_\tau^D = D(D + \tau K)^{-1}D$  is symmetric positive definite, the lumped energy  $E_\tau^D$  is concave, its linearisation is the objective the assignment already maximises, and with an exact assignment Proposition 4.1 applies verbatim. Mass lumping is a standard, first-order-consistent variant of the P1 heat step [10]; Propositions 5.1–5.4 hold unchanged with  $L = D^{-1}K$ . This changes the diffusion operator and therefore every measured number, so it is a separate experiment with its own falsifiers; it is recorded here because it is the one-line change that removes the skew term of §7.2(1) exactly — with  $W = D(D + \tau K)^{-1}D$ , which is symmetric positive definite, the lumped energy is the graph heat content of [18, §5.2] and the assignment objective is its linearisation, so Proposition 4.1 applies — leaving only the inexactness of §7.2(2). Since the skew term is  $\lesssim 2 \times 10^{-3}$  in matrix norm and  $\sim 10^{-5}$  at an actual labelling pair, the experiment would be isolating a small effect.

## 8 The time step: a two-sided window

### 8.1 The rule

Per mesh level,

$$\tau = \min((c h_{\text{mean}})^2, (\rho R_{\text{cell}})^2), \quad c = 4, \quad \rho = 1, \quad (22)$$

so  $\sqrt{\tau}$  is  $c$  mean edges unless that would exceed the cell radius. The two branches are the two sides of the window; the constants are pre-registered and the measured band that passes every gate is in report 08 and `CLAUDE.md`.

## 8.2 Freeze

A vertex's label can change only if the diffused field crosses its neighbour's at that vertex, which requires the continuum interface to move at least a mesh spacing. With Proposition 5.4 and  $\kappa \sim 1/R_{\text{cell}}$  for a cell-scale feature,

$$\delta = \frac{\tau}{2R_{\text{cell}}} \gtrsim h \iff \tau \gtrsim 2hR_{\text{cell}} \iff c \gtrsim \sqrt{2R_{\text{cell}}/h} \quad \text{on the uncapped branch.} \quad (23)$$

The implementation reports the same criterion without the factor 2 (it was written with the Gaussian displacement  $\tau\kappa$ ): `c_lo_hmean` =  $\sqrt{R_{\text{cell}}/h_{\text{mean}}}$  and `c_lo_hmax_local` =  $\sqrt{R_{\text{cell}}/h_{\text{max}}}$ . The factor shifts the threshold by  $\sqrt{2}$ , inside the margin the constants were chosen with (the frozen  $c = 2$  / converged  $c = 4$  pair is the NC3-CAL control of `testing/test_mbo_auction.py`; report 08 §3 cites [28] for the freeze regime); it is noted, not corrected, because the ratio is a diagnostic and every measured run quotes the reported value. Both are order-of-magnitude statements: the expansion behind Proposition 5.4 assumes  $h \ll \sqrt{\tau} \ll R_{\text{cell}}$ , and at the coarsest levels neither inequality is strong. The rigorous statement in this direction is for graphs: [28, Theorem 4.2] proves the two-phase graph scheme is stationary for every initial set when  $\tau < \log(3/2)/\rho(\Delta) \approx 0.4/\rho(\Delta)$ ,  $\rho$  the spectral radius. For a mesh Laplacian  $\rho(L) \sim C/h^2$ , so that bound is  $\sqrt{\tau} \lesssim 0.6h/\sqrt{C}$  — a much weaker (necessary-for-motion) condition than (23), which involves  $R_{\text{cell}}$ .

## 8.3 Over-merge

On the other side, when  $\sqrt{\tau}$  approaches  $R_{\text{cell}}$  a cell's own diffused indicator no longer dominates at its own centre. From the mixture, for a disc of radius  $R$  the diffused indicator at the centre is

$$y(0) = \int_0^\infty \frac{e^{-t/\tau}}{\tau} (1 - e^{-R^2/4t}) dt = 1 - x K_1(x), \quad x = \frac{R}{\sqrt{\tau}}, \quad (24)$$

( $K_1$  the modified Bessel function; the Gaussian value is  $1 - e^{-x^2/4}$ ). At the cap  $\rho = 1$ ,  $x = 1$  and  $y(0) = 1 - K_1(1) = 0.398$ : the cell's own signal at its core is below  $\frac{1}{2}$ , i.e. the *sum* of its neighbours' diffused indicators already exceeds it there. The *argmax* compares against the *largest* neighbour, so this does not by itself imply core loss; it marks the scale at which core loss becomes possible. At  $x = 2$  and  $4$ ,  $y(0) = 0.720$  and  $0.950$ . This is why  $\rho = 1$  is called the *loosest defensible* cap: it is the point at which the diffusion has spread over the whole cell, and anything tighter is a preference. Assignment quality cannot see over-merging — the balanced threshold hits its area target by construction whatever the geometry — so the over-merge instruments are geometric (§11).

## 8.4 The reported ratios, and anisotropy

`tau_diagnostics` reports  $\sqrt{\tau}/h_{\text{mean}}$ ,  $\sqrt{\tau}/h_{\text{max}}$ ,  $\sqrt{\tau}/R_{\text{cell}}$ , the two freeze constants, `cap_active`, and the *freeze ratio*

$$\frac{\sqrt{\tau}/h_{\text{max}}}{\sqrt{R_{\text{cell}}/h_{\text{max}}}} = \sqrt{\frac{\tau}{R_{\text{cell}} h_{\text{max}}}} = \begin{cases} c h_{\text{mean}}/\sqrt{R_{\text{cell}} h_{\text{max}}} & \text{uncapped,} \\ \sqrt{R_{\text{cell}}/h_{\text{max}}} & \text{capped,} \end{cases} \quad (25)$$

which is the quantity `CLAUDE.md` calls the  $\tau$  freeze ratio. Two consequences follow from (25) and are borne out by the runs. On the capped branch the ratio *falls* as cells shrink relative to the

mesh ( $R_{\text{cell}} \propto N^{-1/2}$ ), which is why the standard  $100 \times 96$  base level drops below 1 at  $N = 750$  and 1000; on the uncapped branch it falls as cells *grow* relative to the mesh, which is why a well-resolved low- $N$  configuration runs at 0.5–0.6 and passes every gate — the ratio is not a defect indicator on its own, and the measured band is the reference. The mesh is anisotropic ( $h_{\text{max}}/h_{\text{mean}} = 1.81$  on the torus ladder), so a mean-based figure flatters the coarse band: the rule is on  $h_{\text{mean}}$  and the freeze ratio is on  $h_{\text{max}}$ , deliberately.

## 8.5 The continuation probe

$E_\tau$  depends on the labels alone once  $\tau$  is fixed, so a level that froze and a level that converged both show a flat energy tail. The probe `tau_continuation_probe` re-runs one balanced threshold with  $\tau' = f\tau$ ,  $f \in \{2, 4\}$ , and evaluates the resulting labels at the *level's*  $\tau$ : if a wider blur reaches a labelling with materially lower  $E_\tau$  and moves a material fraction of vertices, the level was pinned. Because a wider blur always re-tiebreaks some boundary vertices and shaves some energy, the verdict needs calibrated thresholds (`nc3_K`, `nc3_f_min`), and until they are pinned by the NC3-CAL control the probe records and reports `pinned = None` rather than a verdict. A pinned level would be annealed ( $\tau \leftarrow 1.5\tau$ , at most three times) and re-run. With the shipped defaults `nc3_K = nc3_f_min = None` the verdict is always `None`, so the anneal branch is unreachable and every measured run reports `n_anneals = 0` vacuously — the probe records, it does not yet decide.

Code: the window

```
src/partition/mbo-auction.py: tau_diagnostics(mesh, n_cells, cfg, tau_override) implements (22) and reports every ratio of §8.4; tau_continuation_probe is §8.5; MBOConfig.tau_c, rho, anneal_factor, max_anneals, probe_factors, nc3_K, nc3_f_min. Only tau_c, rho and max_iters are config-reachable (mbo: block); the rest define the measurement protocol.
```

## 9 Initialisation

The first labelling is one balanced assignment of geodesic scores:

1.  $N$  sites by greedy farthest-point sampling in the *ambient* Euclidean metric of  $\mathbb{R}^3$  (first site from the seeded RNG, each next site the vertex farthest from the current set) — `farthest_point_sampling`, shared with the PGD seeded init so both start from the same site layout;
2.  $d_{ik}$  = shortest-path distance from site  $k$  to vertex  $i$  on the edge graph of the mesh, edge weights the Euclidean edge lengths, by  $N$  single-source Dijkstra searches returning the full  $V \times N$  matrix (`dijkstra` with `indices=sites`) — a graph metric that *over*-estimates the geodesic distance (paths are confined to edges) but respects the surface topology, since it is built from `faces`; the heat method of [7] is not what is implemented;
3. scores  $s_{ik} = -d_{ik}^2$  and one full-budget balanced assignment.

With  $-d^2$  scores, (18) is a *power diagram* of the graph metric with weights  $\psi$  (doc 09, §8; [3]): the capacity-constrained Voronoi diagram that [6] also use as an initialisation, computed here by the transportation dual rather than by their gradient method, and on a surface. It is also, by construction, step 0 of approach C (capacity-constrained geodesic Lloyd), which is what makes the attribution control of report 08 possible: running the init alone on the finest mesh isolates what the dynamics add.

The balanced start is a **hard requirement**, measured rather than argued: from a cold unbalanced Voronoi labelling the first balanced threshold leaves a worst cell at 68.16% area deviation, from the

balanced one at 1.97% (docs/plans/PHASE1\_BC\_REPLACEMENT\_PLAN.md, Phase A, and the module docstring; report 08 does not carry these two numbers).

#### Code: initialisation

```
src/partition/mbo_auction.py:          geodesic_balanced_init(mesh, n_cells,
seed);          edge_graph(mesh);          src/optimization/initialization.py:
farthest_point_sampling(vertices, n_seeds, seed); scipy.sparse.csgraph.dijkstra.
```

## 10 The mesh ladder

Levels are the same mesh ladder Phase 1 uses. Within a level, steps repeat until the fraction of vertices that change label is below `churn.tol` =  $10^{-4}$  for `patience` = 3 consecutive steps, or `max_iters` = 200 is hit (never, in the measured runs: the most any level took is 66). Between levels the labelling is carried by *nearest embedded vertex*: each fine vertex takes the label of the nearest coarse vertex in  $\mathbb{R}^3$  (`cKDTree`). Labels are interpolated rather than densities, because a hard partition has no density to interpolate and the nearest-vertex map of a labelling is itself a labelling. What the transfer preserves: each cell's territory up to a band of width  $O(h_{\text{coarse}})$  along its boundary, hence its area up to  $O(h_{\text{coarse}} \cdot \text{Per}(\Omega_k))$  — the first balanced threshold on the fine level removes that error. What it does not preserve: connectivity is not guaranteed across the transfer (a thin neck can be cut), and  $\tau$  is recomputed from (22) on the new mesh, so the effective diffusion length shrinks with  $h$  on the uncapped branch. Report 08 records fragmented cells born at one level and healed at the next; the ladder is what heals them.

#### Code: ladder

```
src/partition/mbo_auction.py:          run_mbo_ladder(meshes, n_cells, cfg, init_only),
mbo_level(mesh, labels, n_cells, cfg, level, run_probe),          run_steps,
interpolate_labels(old.vertices, new.vertices, labels).
```

## 11 Instruments

None of these is a gate on the dynamics; they are what the reports quote.

- **Energies.**  $E_\tau^D$  (`E_lumped`, primary because its linearisation differs from the assignment objective only by the skew term of §7.2) and  $E_\tau^M$  (`E_consistent`), both by (20) with the respective weight, and both  $\rightarrow \frac{1}{2} \sum_k \text{Per}(\Omega_k)$  by Proposition 5.3. Descriptive trends only (§7.2).
- **Label boundary length.** The midpoint-cut construction: a triangle with two labels contributes the segment joining the midpoints of its two bi-labelled edges, one with three labels contributes three segments from those midpoints to the centroid (`label_boundary_length`, `src/partition/balanced_readout.py`). It counts each interface once; the Phase 2 perimeter counts each twice, and the measured ratio between them on an unoptimised labelling is 1.999 (`CLAUDE.md`).
- **Compactness.**  $Q_k = P_k^2 / (4\pi A_k)$  with  $P_k$  the per-cell boundary length (same construction, each segment credited to both cells) and  $A_k = T_k$ ;  $Q = 1$  is a disc. Over-merging raises  $Q$ .
- **Core loss.** The number of cells whose own diffused-score peak vertex,  $\arg \max_i y_{ik}$ , is not in their territory: the most direct over-merge symptom, since it says the cell's signal was swamped before the threshold was taken.

- **The three Phase 1 gates**, run on every level's labels through the same harness path as the final labels (`run_gates`). For one-hot labels the dormant gate is vacuous and the area gate is the solver check of §6.3; only connectivity has content, because nothing in §6 enforces it.

Code: instruments

```
src/partition/mbo_auction.py:      lyapunov_energy,      per_cell_boundary_lengths,
compactness, core_loss; src/partition/balanced_readout.py:  label_boundary_length;
src/partition/arm_harness.py: run_gates, labels_to_one_hot.
```

## Summary table

| Quantity                                                                              | Status                                                                                                                              | Implementation                                                                     |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Diffusion step $(M + \tau K)y_k = M\chi_k$                                            | derived (backward Euler of the P1 heat equation)                                                                                    | <code>make_diffusion_solver</code> , <code>diffuse_indicators</code>               |
| Mixture representation (10)                                                           | proved                                                                                                                              | — (analysis)                                                                       |
| $\sum_k y_k = \mathbf{1}$ ; $MA_\tau$ SPD; no discrete positivity                     | proved                                                                                                                              | —                                                                                  |
| Continuum kernel $\frac{1}{2\pi\tau}K_0(r/\sqrt{\tau})$ , marginal (12)               | derived                                                                                                                             | —                                                                                  |
| Interface constant $c_{K_\tau^{\text{res}}} = \frac{1}{2}$ (Gaussian $1/\sqrt{\pi}$ ) | proved; $\Gamma$ -limit cited [13] (flat)                                                                                           | <code>lyapunov_energy</code> approximates $\frac{1}{2}\sum_k \text{Per}(\Omega_k)$ |
| Per-step displacement $\tau\kappa/2$                                                  | proved (leading order); exact disc solution (16); mesh at 0.91–0.97 of it and 0.48–0.54 of the Gaussian, placement-averaged         | §5.6                                                                               |
| Balanced threshold $\arg \max_k (y_{ik} + \psi_k)$                                    | derived (transportation dual, doc 09); $N=2$ case is Ruuth–Wetton                                                                   | <code>balanced_assign</code> <code>solve_dual_offsets</code> →                     |
| Dissipation with exact assignment                                                     | theorem (cited); the implemented step is the exact MM step for $E_\tau^D$ up to a $\lesssim 10^{-3}$ skew term and the inexact dual | —                                                                                  |
| Descent inequality (21)                                                               | proved for any argmax labelling                                                                                                     | <code>descent_slack</code> ; gate G4                                               |
| Lumped–lumped pairing restores the theorem                                            | stated; <b>not implemented</b>                                                                                                      | —                                                                                  |
| $\tau$ rule (22); freeze (23); over-merge (24)                                        | derived (order of magnitude; $y(0) = 1 - xK_1(x)$ exact for the continuum kernel)                                                   | <code>tau_diagnostics</code>                                                       |
| Freeze ratio (25) and its two regimes                                                 | derived                                                                                                                             | <code>tau_diagnostics</code>                                                       |
| Continuation probe                                                                    | protocol; verdict uncanceled until NC3-CAL                                                                                          | <code>tau_continuation_probe</code>                                                |
| Init as power diagram of the graph metric                                             | derived                                                                                                                             | <code>geodesic_balanced_init</code>                                                |
| Ladder transfer error $O(h \text{Per})$ $E_\tau$ , boundary length, $Q_k$ , core loss | stated instruments                                                                                                                  | <code>interpolate_labels</code> §11                                                |

## Related documents

- Measurements, pre-registration and the negative controls:  
docs/experiments/08-mbo-auction-dynamics/ and  
docs/plans/PHASE1\_BC-REPLACEMENT\_PLAN.md (Phase A).
- The transportation dual and the readout it was built for:  
docs/math/09-balanced-readout/.
- The P1 discretisation and the Phase 1 energy this replaces:  
docs/math/06-phase1-energy-discretization/.
- Taxonomy of the high- $N$  approaches:  
docs/reference/PHASE1\_HIGHN\_APPROACHES\_ABCDE.md.
- Held copies of every cited paper: docs/papers/README.md.

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